(2) + (3) # Trigonometry 1

Paper 2

BASIC + RULES

QUESTION 1

1.1. Given: $\sin\alpha = \frac{3}{5}$ and $90° < \alpha < 270°$

With the aid of a sketch and **without using a calculator**, determine:

1.1.1. $\tan\alpha$ (3)

1.1.2. $\cos(90+\alpha)$ (2)

1.1.3. $\cos 2\alpha$ (3)

1.2. Given: $\sin 38° = p$, determine the following in terms of p.
(Without using a calculator.)

1.2.1. $\cos(-38°)$ (3)

1.2.2. $\sin 76°$ (2)

1.3. Simplify the following without using a calculator:

$$\frac{\sin 150°.\tan 225°}{\sin(-30°).\sin 420°}$$ (6)

1.4. Prove:

$$\frac{2\sin^2 x}{2\tan x - \sin 2x} = \frac{1}{\tan x}$$ (6)

QUESTION 2

2.1. If $\cos 62° = m$, determine the value of each of the following in terms of m.

2.1.1. $\sin 28°$ (2)

2.1.2. $\cos 362°$ (4)

2.2. Simplify to a single ratio:

$$\frac{\tan(360° - x).\sin\ (90^0 + x)}{\sin(-x)}$$ (5)

2.3. If $4\sin^2\theta - 3 = 0,\ 90° \le \theta \le 180°$, calculate the value of:

$$\cos\frac{1}{4}\theta.\sin\frac{1}{2}\theta - \tan(3\theta - 45°)$$ (6)

QUESTION 3

Prove that $\frac{\cos 2x}{1+\sin 2x} = \frac{\cos x - \sin x}{\sin x + \cos x}$ [4]

QUESTION 4

4.1. Simplify without using a calculator $\dfrac{\cos 195°.\cos 285°}{\cos(60°-x)\cos x - \sin(60°-x)\sin x}$ (8)

4.2. Prove : $\dfrac{\cos(x-360°) - \tan(180°-x)\sin(360°-2x)\cos x}{\sin(90°+x)} = \cos 2x$ (7)

4.3. Prove: $\dfrac{\sin x \cos x}{\frac{1}{2}\cos^2 x - \frac{1}{2}\sin^2 x} = \tan 2x$ (3)

4.4. Determine for which values of x, $x \in [0°; 360°]$, is the identity in 5.3.1. undefined? (3)

4.5. Prove that $\dfrac{\sin 5x + \sin 3x}{\cos 5x - \cos 3x} = \dfrac{-1}{\tan x}$

TRIG EQUATIONS

QUESTION 5

5.1. Determine the general solution of

5.1.1. $2 \sin 3\theta + 1 = 0$

5.1.2. $\tan 2\theta = -\tan(\theta + 45°)$

5.1.3. $\cos 2\theta - \sin^2\theta - \cos\theta = 0$

5.1.4. $8 \sin^2\theta + 7 \sin 2\theta + 4 = 0$

5.1.5. $\tan x \sin x + \cos x \tan x = 0$

5.2. Solve for α : $2\cos(\alpha + 15°) - 1 = 0$; $x \in [-180°; 180°]$.

5.3. Solve for x : $\cos(x + 30°) = \sin 3x$; $x \in [0°; 360°]$.

INTERESTING

QUESTION 6

Determine the maximumvalue of $3 \sin\alpha \cos 20° - 3\cos\alpha \sin 200°$

QUESTION 7

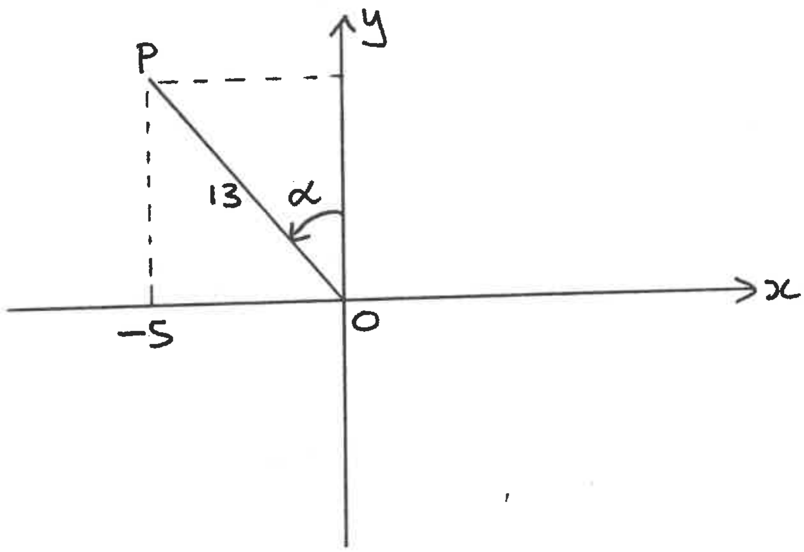
7.1. Determine the values of the following, without the use of a calculator:

7.1.1. $$\frac{\sin 160^\circ \cos(-20^\circ)}{\cos 13^\circ \cos 37^\circ + \sin(-13^\circ)\sin 37^\circ}$$

7.1.2. $$\sin^2 10^\circ + \sin^2 80^\circ - \frac{\sin^2 22{,}5^\circ}{\cos 300^\circ}$$

7.1.3. $$(\cos 15^\circ + \sin 15^\circ)^2$$

7.2. Refer to the diagram:



$x_P = -5$

$OP = 13$ units

Determine the value of

7.2.1. $\cos(90^\circ + \alpha)$

7.2.2. $\sin(90^\circ + \alpha)$

7.2.3. $\sin \alpha$

7.2.4. $\tan \alpha$

7.2.5. $\sin 2\alpha$

7.2.6. $\cos(\alpha + 30^\circ)$

7.3. If $\sin 11^\circ = p$, determine the following in terms of p:

7.3.1. $\sin 191^\circ$ 7.3.2. $\cos 101^\circ$

7.3.3. $\cos 11^\circ$ 7.3.4. $\tan 11^\circ$

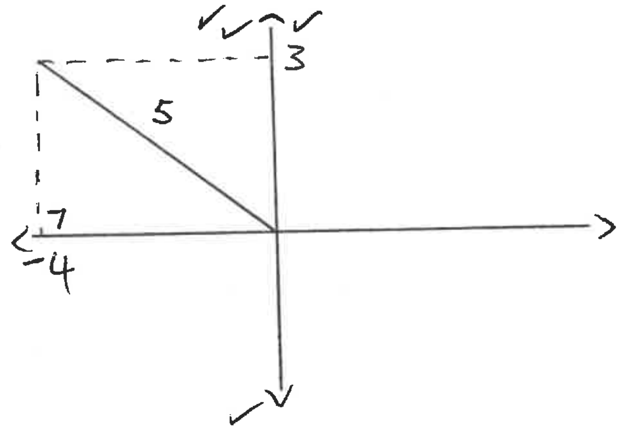
7.3.5. $\sin 22^\circ$ 7.3.6. $\tan 22^\circ$ 7.3.7. $\cos 41^\circ$

Trigonometry 1: Solutions

Basics and Rules

Question 1:

1.1.



$x^2 + y^2 = r^2$

$x^2 + 3^2 = 5^2$

$x^2 = 16$

$x = -4$

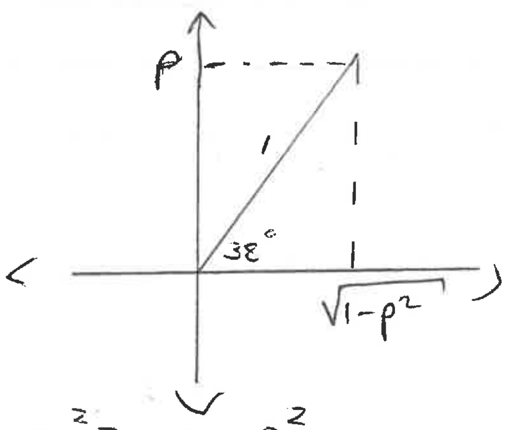
1.1.1. $\tan\alpha = y/x = \frac{3}{-4} = -\frac{3}{4}$

1.1.2. $\cos(90 + \alpha) = -\sin\alpha = -3/5$

1.1.3. $\cos 2\alpha = 1 - 2\sin^2\alpha$

$= 1 - 2(3/5)^2$

$= 7/25$



1.2.1. $\cos(-38°)$

$= \cos 38°$

$= \sqrt{1-p^2}$

1.2.2. $\sin 76°$

$= 2\sin 38° \cos 38°$

$= 2p \cdot \sqrt{1-p^2}$

$x^2 = 1 - p^2$

$x = \sqrt{1-p^2}$

1.3. $\dfrac{\sin 150^\circ \tan 225^\circ}{\sin(-30^\circ).\ \sin 420^\circ}$

$= \dfrac{\sin(180^\circ - 30^\circ).\ \tan(180^\circ + 45^\circ)}{-\sin 30^\circ.\ \sin(360^\circ + 60^\circ)}$

$= \dfrac{\sin 30^\circ.\ \tan 45^\circ}{-\sin 30^\circ.\ \sin 60^\circ}$

$= \dfrac{\frac{1}{2}.\ (1)}{-(\frac{1}{2})\ (\sqrt{3}/2)}$

$= -\dfrac{2}{\sqrt{3}}$

$= -\dfrac{2\sqrt{3}}{3}$

1.4. LHS $\dfrac{2\sin^2 x}{2\tan x - \sin 2x}$ RHS $\dfrac{1}{\tan x}$

$= 2\sin^2 x \div \left(2\tan x - \sin 2x\right)$ $= \dfrac{\cos x}{\sin x}$

$= 2\sin^2 x \div \left(\dfrac{2\sin x}{\cos x} - 2\sin x.\cos x\right)$

$= 2\sin^2 x \div \left(\dfrac{2\sin x - 2\sin x.\cos^2 x}{\cos x}\right)$

$= 2\sin^2 x \div \left(\dfrac{2\sin x(1 - \cos^2 x)}{\cos x}\right)$

$= 2\sin^2 x \div \left(\dfrac{2\sin x.\ \sin^2 x}{\cos x}\right)$

$= 2\sin^2 x \times \left(\dfrac{\cos x}{2\sin x.\ \sin^2 x}\right)$

$= \dfrac{\cos x}{\sin x}$

$\therefore$ LHS = RHS

Question 2:

2.1.1. $\sin 28^\circ$

$= \sin(90^\circ - 62^\circ)$

$= \cos 62^\circ$

$= \underline{m}$

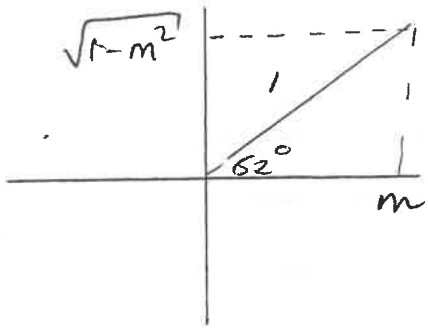
2.1.2. $\cos 362^\circ$

$= \cos(300^\circ + 62^\circ)$

$= \cos 300^\circ . \cos 62^\circ - \sin 300^\circ \sin 62^\circ$

$= \cos(360^\circ - 60^\circ)\cos 62^\circ - \sin(360^\circ - 60^\circ)\sin 62^\circ$

$= \cos 60^\circ \cos 62^\circ + \sin 60^\circ \sin 62^\circ$



$\therefore y = \sqrt{1-m^2}$

$= \underline{\frac{1}{2} m + \frac{\sqrt{3}}{2}\sqrt{1-m^2}}$

2.2. $\frac{\tan(360^\circ - x).\sin(90^\circ + x)}{\sin(-x)}$

$= \frac{-\tan x . \cos x}{-\sin x}$

$= -\frac{\sin x}{\cos x} . \frac{\cos x}{-\sin x}$

$= \underline{1}$

2.3 $4\sin^2\theta - 3 = 0$

$\sin^2\theta = 3/4$

$\sin\theta = \pm\sqrt{3}/2$

$\therefore \theta = 180° - 60°$ ($\angle$ in $\underline{II}$)

$= 120°$

$\cos\frac{1}{4}(120°) . \sin\frac{1}{2}(120°) - \tan(3(120°) - 45°)$

$= \cos 30° . \sin 60° - \tan 315°$

$= \cos 30° . \sin 60° - \tan(360° - 45°)$

$= \cos 30° . \sin 60° + \tan 45°$

$= \sqrt{3}/2 . \sqrt{3}/2 + 1$

$= 3/4 + 1$

$= \underline{7/4}$

Question 3

LHS $\dfrac{\cos 2x}{1+\sin 2x}$

$= \dfrac{\cos^2 x - \sin^2 x}{1 + 2\sin x \cos x}$

$= \dfrac{\cos^2 x - \sin^2 x}{\cos^2 x + \sin^2 x - 2\sin x \cos x}$

$= \dfrac{\cos^2 x - \sin^2 x}{\cos^2 x - 2\sin x \cos x + \sin^2 x}$

$= \dfrac{(\cos x - \sin x)(\cos x + \sin x)}{(\cos x + \sin x)^2}$

$= \dfrac{\cos x - \sin x}{\cos x + \sin x}$

$=$ RHS

Question 4

4.1. $\dfrac{\cos 195^\circ \cos 285^\circ}{\cos(60^\circ - x).\cos x - \sin(60^\circ - x)\sin x}$

$= \dfrac{\cos(180^\circ + 15^\circ)\cos(360^\circ - 75^\circ)}{\cos(60^\circ - x + x)}$

$= \dfrac{-\cos 15^\circ \quad \cos 75^\circ}{\cos 60^\circ}$

$= \dfrac{-\cos 15^\circ \quad \sin 15^\circ}{1/2}$

$= -2\sin 15^\circ \cos 15^\circ$

$= -2\sin 30^\circ$

$= -2\left(\frac{1}{2}\right)$

$= -1$

4.2. $\dfrac{\cos(x-360^\circ) - \tan(180^\circ - x)\sin(360^\circ - 2x).\cos x}{\sin(90^\circ + x)}$

$= \dfrac{\cos(-(360^\circ - x)) - (-\tan x).(-\sin 2x).\cos x}{\cos x}$

$= \dfrac{\cos(360^\circ - x) - \tan x.\sin 2x.\cos x}{\cos x}$

$= \dfrac{\cos x - \tan x.\sin 2x.\cos x}{\cos x}$

$= \dfrac{\cos x(1 - \tan x.\sin 2x)}{\cos x}$

$= 1 - \dfrac{\sin x}{\cos x} . 2\sin x.\cos x$

$= 1 - 2\sin^2 x$

$= \underline{\cos 2x}$

4.3. $\dfrac{\sin x \cos x}{\frac{1}{2}\cos^2 x - \frac{1}{2}\sin^2 x}$

$= \dfrac{\sin x \cos x}{\frac{1}{2}(\cos^2 x - \sin^2 x)}$

$= \dfrac{\sin x \cos x}{\frac{1}{2} \cos 2x}$

$= \dfrac{2\sin x \cos x}{\cos 2x}$

$= \dfrac{\sin 2x}{\cos 2x}$

$= \underline{\tan 2x}$

4.4. $\frac{1}{2}\cos 2x = 0$

$\cos 2x = 0$

$2x = 90^\circ + 180k; \quad k \in \mathbb{Z}$

$\underline{x = 45^\circ + 90k}$

$\tan 2x$ undefined

$2x = 90^\circ + 180^\circ k, \quad k \in \mathbb{Z}$

$\underline{x = 45^\circ + 90^\circ k}$

4.5. LHS $\dfrac{\sin 5x + \sin 3x}{\cos 5x - \cos 3x}$

$= \dfrac{\sin(4x+x) + \sin(4x-x)}{\cos(4x+x) - \cos(4x-x)}$

$= \dfrac{\sin 4x\cos x + \cos 4x\sin x + \sin 4x\cos x - \cos 4x\sin x}{\cos 4x\cos x - \sin 4x\sin x - (\cos 4x.\cos x + \sin 4x\sin x)}$

$= \dfrac{2\sin 4x\cos x}{\cos 4x\cos x - \sin 4x\sin x - \cos 4x\cos x - \sin 4x.\sin x}$

$= \dfrac{2\sin 4x\cos x}{-2\sin 4x\sin x}$

$= -\dfrac{\cos x}{\sin x}$

RHS $-1 \div \tan x$

$= -1 \div \dfrac{\sin x}{\cos x}$

$= -1 \times \dfrac{\cos x}{\sin x}$

$= \dfrac{-\cos x}{\sin x}$

$\therefore$ LHS = RHS

Trig Equations

Question 5

5.1.1. $2\sin 3\theta + 1 = 0$

$\sin 3\theta = -\frac{1}{2}$

ref ∠ $\sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$

III

$3\theta = 180^\circ + 30^\circ + 360k;\ k \in \mathbb{Z}$

$3\theta = 210^\circ + 360k$

$\theta = 70^\circ + 120k$

IV

$3\theta = 360^\circ - 30^\circ + 360k;\ k \in \mathbb{Z}$

$3\theta = 330^\circ + 360k$

$\theta = 110^\circ + 120k$

5.1.2. $\tan 2\theta = -\tan(\theta + 45^\circ)$

$2\theta = 180^\circ - (\theta + 45^\circ) + 180^\circ k;\ k \in \mathbb{Z}$

$2\theta = 180^\circ - \theta - 45^\circ + 180^\circ k$

$3\theta = 135^\circ + 180^\circ k$

$\theta = 45^\circ + 60^\circ k$

5.1.3

$$\cos 2\theta - \sin^2\theta - \cos\theta = 0$$
$$2\cos^2\theta - 1 - (1-\cos^2\theta) - \cos\theta = 0$$
$$2\cos^2\theta - 1 - 1 + \cos^2\theta - \cos\theta = 0$$
$$3\cos^2\theta - \cos\theta - 2 = 0$$
$$(3\cos\theta + 2)(\cos\theta - 1) = 0$$

$\cos\theta = -2/3$ OR $\cos\theta = 1$

ref $\angle = \cos^{-1}(2/3)$ $\theta = 0° + 360k;\ k\in\mathbb{Z}$

$= 48,19°$

II: $\theta = 180° - 48,19° + 360k.\ k\in\mathbb{Z}$
$= \underline{131,81° + 360k}$

III: $\theta = 180° + 48,19° + 360k.\ k\in\mathbb{Z}$
$= \underline{228,19° + 360k}$

5.1.4

$$8\sin^2\theta + 7\sin 2\theta + 4 = 0$$
$$8\sin^2\theta + 7(2\sin\theta\cos\theta) + 4(\sin^2\theta + \cos^2\theta) = 0$$
$$8\sin^2\theta + 14\sin\theta\cos\theta + 4\sin^2\theta + 4\cos^2\theta = 0$$
$$12\sin^2\theta + 14\sin\theta\cos\theta + 4\cos^2\theta = 0$$
$$6\sin^2\theta + 7\sin\theta\cos\theta + 2\cos^2\theta = 0$$
$$(3\sin\theta + 2\cos\theta)(2\sin\theta + \cos\theta) = 0$$

$3\sin\theta = -2\cos\theta$
$\tan\theta = -2/3$

$\theta = 180° - 33,69° + 180k;\ k\in\mathbb{Z}$

$= \underline{146,31° + 180k}$

$2\sin\theta = -\cos\theta$
$\tan\theta = -1/2$

$\theta = 180° - 26,57° + 180k;$
$k\in\mathbb{Z}$

$\theta = \underline{153,43° + 180k}$

10

5.1.5 $\tan x \sin x + \cos x \tan x = 0$

$\tan x \ (\sin x + \cos x) = 0$

$\tan x = 0$ OR $\sin x - \cos x = 0$

$x = 0 + 180k, \ k \in \mathbb{Z}$

$\sin x = \cos x$

$\frac{\sin x}{\cos x} = 1$

$\tan x = 1$

$x = 45^\circ + 180k; \ k \in \mathbb{Z}$

5.2. $2\cos(\alpha+15^\circ) - 1 = 0$

$$\cos(\alpha+15^\circ) = \tfrac{1}{2}$$

$$\text{ref} \angle = 60^\circ$$

$\underline{I}$

$$\alpha + 15^\circ = 60^\circ + 360^\circ k;\ k \in \mathbb{Z}$$
$$\alpha = 45^\circ + 360^\circ k$$

$\underline{IV}$

$$\alpha + 15^\circ = 360^\circ - 60^\circ + 360 k;\ k \in \mathbb{Z}$$
$$\alpha = 285^\circ + 360 k$$

$$\therefore \alpha \in \{-75^\circ; 45^\circ\}$$

5.3. $\cos(x+30^\circ) = \sin 3x$

$$\cos(x+30^\circ) = \cos(90^\circ - 3x)$$

$\underline{I}$

$$x + 30^\circ = 90^\circ - 3x + 360^\circ k;\ k \in \mathbb{Z}$$
$$4x = 60^\circ + 360^\circ k$$
$$\underline{x = 15^\circ + 90^\circ k},$$

$\underline{IV}$

$$x + 30^\circ = 360^\circ - (90^\circ - 3x) + 360^\circ k;\ k \in \mathbb{Z}$$
$$x + 30^\circ = 270^\circ + 3x + 360^\circ k$$
$$-2x = 240^\circ + 360^\circ k$$
$$\underline{x = 120^\circ - 180^\circ k}$$

$$x \in \{15^\circ; 60^\circ; 105^\circ; 195^\circ; 240^\circ; 285^\circ\}$$

Question 6

$3\sin\alpha\cos 20^\circ - 3\cos\alpha\sin 200^\circ$

$= 3\sin\alpha\cos 20^\circ - 3\cos\alpha\sin(180^\circ + 20^\circ)$

$= 3\sin\alpha\cos 20^\circ + 3\cos\alpha\sin 20^\circ$

$= 3(\sin\alpha\cos 20^\circ + \cos\alpha\sin 20^\circ)$

$= 3(\sin(\alpha + 20^\circ))$

$= 3\sin(\alpha + 20^\circ)$

max value of sin is 1.

$\therefore$ max value of $3\sin(\alpha+20^\circ)$ is $\underline{3}$

Question 7:

7.1.1. $\dfrac{\sin 160^\circ \cos(-20^\circ)}{\cos 13^\circ \cos 37^\circ + \sin(-13^\circ)\sin 37^\circ}$

$= \dfrac{\sin(180^\circ - 20^\circ)\cos 20^\circ}{\cos 13^\circ \cos 37^\circ - \sin 13^\circ \sin 37^\circ}$

$= \dfrac{\sin 20^\circ \cos 20^\circ}{\cos(13^\circ + 37^\circ)}$

$= \dfrac{\frac{1}{2}\sin 40^\circ}{\cos 50^\circ}$

$= \dfrac{\frac{1}{2}\cos 50^\circ}{\cos 50^\circ}$

$= \underline{\frac{1}{2}}$

7.1.2. $\sin^2 10 + \sin^2 80 - \dfrac{\sin^2 22{,}5}{\cos 300}$

$= \sin^2 10 + \cos^2 10 - \dfrac{\sin^2 22{,}5}{\cos(360^\circ - 60^\circ)}$

$= 1 - \dfrac{\sin^2 22{,}5}{\frac{1}{2}}$

$= 1 - 2\sin^2 22{,}5$

$= \cos 45$

$= \dfrac{\sqrt{2}}{2}$

7.1.3. $(\cos 15^\circ + \sin 15^\circ)^2$

$= \cos^2 15^\circ + 2\sin 15^\circ \cos 15^\circ + \sin^2 15^\circ$

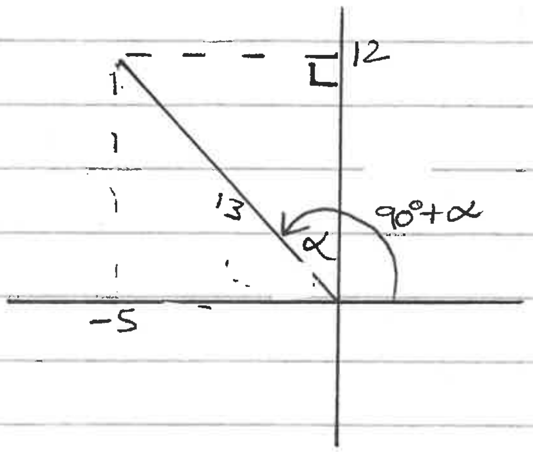
$= 1 + \sin 30^\circ$

$= 1 + \frac{1}{2}$

$= \frac{3}{2}$

7.2. $x^2 + y^2 = r^2$

$y = 12$



7.2.1. $\cos(90^\circ + \alpha)$

$= \dfrac{-5}{13}$

$=$

7.2.2. $\sin(90^\circ + \alpha)$

$= \frac{12}{13}$

7.2.3 $\sin\alpha$

$= \frac{5}{13}$

7.2.4 $\tan\alpha$

$= \frac{5}{12}$

7.2.5 $\sin 2\alpha$

$= 2\sin\alpha\cos\alpha$

$= 2\left(\frac{5}{13}\right)\left(\frac{12}{13}\right)$

$= \frac{120}{169}$

7.2.6. $\cos(\alpha + 30^\circ)$

$= \cos\alpha \cdot \cos 30^\circ - \sin\alpha \cdot \sin 30^\circ$

$= \frac{12}{13} \cdot \frac{\sqrt{3}}{2} - \left(\frac{5}{13}\right)\left(-\frac{1}{2}\right)$

$= \frac{6\sqrt{3}}{13} + \frac{5}{26}$

$= \frac{12\sqrt{3} + 5}{26}$

7.3.1. $\sin 191^\circ$

$= \sin(180^\circ + 11^\circ)$

$= -\sin 11^\circ$

$= -p$

7.3.2 $\cos 101^\circ$

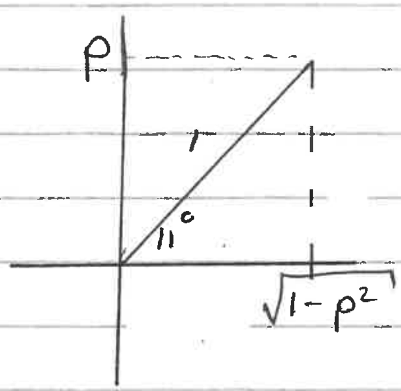
$= \cos(90^\circ + 11^\circ)$

$= -\sin 11^\circ$

$= -p$

7.3.3 $\cos 11^\circ$

$= \sqrt{1-p^2}$



7.3.4 $\tan 11^\circ$

$= \dfrac{p}{\sqrt{1-p^2}}$

7.3.5 $\sin 22^\circ$

$= 2\sin 11^\circ \cdot \cos 11^\circ$

$= 2p \cdot \sqrt{1-p^2}$

7.3.6 $\tan 22^\circ$

$= \dfrac{\sin 22^\circ}{\cos 22^\circ}$

$= \dfrac{2p\sqrt{1-p^2}}{1 - 2\sin^2 11^\circ}$

$= \dfrac{2p\sqrt{1-p^2}}{1-2p^2}$

7.3.7 $\cos 41^\circ$

$= \cos(30^\circ + 11^\circ)$

$= \cos 30^\circ \cdot \cos 11^\circ - \sin 30^\circ \sin 11^\circ$

$= \frac{\sqrt{3}}{2} \cdot \sqrt{1-p^2} - \frac{1}{2} p$